



UNSW Course Outline

BABS1201 Molecules, Cells and Genes - 2025

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General Course Information

Course Code : BABS1201

Year : 2025

Term : Term 3

Teaching Period : T3

Is a multi-term course? : No

Faculty : Faculty of Science

Academic Unit : School of Biotechnology and Biomolecular Sciences

Delivery Mode : In Person

Delivery Format : Standard

Delivery Location : Kensington

Campus : Sydney

Study Level : Postgraduate, Undergraduate

Units of Credit : 6

Useful Links

[Handbook Class Timetable](#)

Course Details & Outcomes

Course Description

This course introduces the cell as the simplest form of life, the elements essential to life and the characteristics of living things. Different cell types are compared including their structures and functions. The structures of proteins, fats and carbohydrates, their metabolism and roles in the

cell including the processes for energy generation in cells. The structure of nucleic acid and its role in the processes of cell division and gene expression, and how this applies to genetic inheritance and evolution. This is a blended course. The weekly laboratory classes are delivered face-to-face and lectures are online.

Course Aims

This course aims to:

1. Provide a foundation for further study in cell and molecular biology, biochemistry and genetics. This is done by introducing cells structures and functions, macromolecules involved in cell processes, and the structure and inheritance of genetic material.
2. Introduce scientific literature and provide opportunities to use this literature and develop science communication skills in written and verbal forms.
3. Provide valuable experience working in a biology laboratory working collaboratively as part of a group.

Relationship to Other Courses

These are no prerequisites for BABS1201. It provides a good foundation for BABS1202, although it is not a prerequisite for that course.

Course Learning Outcomes

Course Learning Outcomes
CLO1 : Communicate findings from scientific literature to different audiences using verbal and written formats.
CLO2 : Identify and compare different cell types and their structures and functions.
CLO3 : Describe the structures of the macromolecules proteins, fats and carbohydrates, and their metabolism including the processes for energy generation in cells.
CLO4 : Describe the structures of genetic material, the processes involved in cell division and gene expression, and how this applies to genetic inheritance and evolution.
CLO5 : Demonstrate a knowledge of the techniques and safe work practices required in a biological laboratory, including use of the light microscope, related calculations and basic molecular biology methods.

Course Learning Outcomes	Assessment Item
CLO1 : Communicate findings from scientific literature to different audiences using verbal and written formats.	• Literature review • Science Communication Project
CLO2 : Identify and compare different cell types and their structures and functions.	• Test 1
CLO3 : Describe the structures of the macromolecules proteins, fats and carbohydrates, and their metabolism including the processes for energy generation in cells.	• Test 1
CLO4 : Describe the structures of genetic material, the processes involved in cell division and gene expression, and how this applies to genetic inheritance and evolution.	• Test 2
CLO5 : Demonstrate a knowledge of the techniques and safe work practices required in a biological laboratory, including use of the light microscope, related calculations and basic molecular biology methods.	• Test 2 • Test 1

Learning and Teaching Technologies

Moodle - Learning Management System | Microsoft Teams | Echo 360

Learning and Teaching in this course

I have a question about the course. Where do I find the answer?

Check to see if your question is already answered in this Course Outline or on the Moodle site. If not, post your question to the Discussion Forum on the BABS1201 Moodle site.

Email BABS1201@unsw.edu.au if your question is of a personal or sensitive nature. Include your full name and student number and send from your UNSW email account. Please ensure you type this address in and do NOT reply to a Teams calendar invitation to ensure your email goes to the correct place.

You are welcome to contact your lecturers in the course with any questions. If they do not provide their contact details on their lecture slides, please ask your coordinators for them.

How do I contact my demonstrator? Can I email them?

Your demonstrators are generally not available out of class time, as they are contracted only for their teaching hours and preparation. While you can contact them via your Microsoft Teams demonstrator group channel, please know they may not respond until they prepare for your next class.

How do I find answers to questions about specific lecture material?

Read through the corresponding lecture notes whilst listening to the lecture audio recording (lecture notes and recordings can be accessed through the BABS1201 Moodle site). The lecturer may have answered your question during the lecture. Refer to the corresponding references; there are copies of the recommended text in the library and available from the UNSW Bookshop. Use the Lightboard revision videos available via Moodle, if applicable. Attend the Q&A session that corresponds to that lecture. Post your question on a discussion forum in the BABS1201 Moodle site. Email your question to the lecturer, letting them know the course and the lecture to which your question refers.

Assessments

Assessment Structure

Assessment Item	Weight	Relevant Dates
Literature review Assessment Format: Individual Short Extension: Yes (3 days)	25%	Due Date: 17/10/2025 05:00 PM
Test 1 Assessment Format: Individual	30%	Due Date: Week 7: 27 October - 02 November
Science Communication Project Assessment Format: Individual	25%	Due Date: Week 9: 10 November - 16 November
Test 2 Assessment Format: Individual	20%	Due Date: Week 10: 17 November - 23 November

Assessment Details

Literature review

Assessment Overview

You will choose from a list of topics related to the course themes. The process for the literature review involves two parts:

1. Search the literature on the topic and choose three scientific articles (two primary articles and a review), documenting the process you undertook to find and select the scientific articles. (15%)
2. Use the articles to write a referenced mini-review on the area. This review will assess your ability to gather, synthesise, and evaluate scientific literature and correctly format your references. (10%)

Support for this assessment is provided as a lecture and online lesson on scientific literature, and an online lesson developed for this course with the UNSW library.

This assignment is due in Week 5. Written feedback and a completed rubric are provided.

Course Learning Outcomes

- CL01 : Communicate findings from scientific literature to different audiences using verbal and written formats.

Detailed Assessment Description

See the Assessment Hub on Moodle for more details including the Instructions, Topic List, and Marking Rubric.

Assessment Length

2000 words maximum

Submission notes

You have one opportunity to submit. Please check the file name and location of your assignment before uploading, to ensure you upload the correct file. Also, check you receive acknowledgement of your submission (via email or Turnitin) and that your document displays correctly once uploaded. You should allow sufficient time to upload your assessment before the deadline for any unforeseen issues.

Assessment information

If Special Consideration is approved an extension will be provided as the outcome.

Assignment submission Turnitin type

This assignment is submitted through Turnitin and students do not see Turnitin similarity reports.

Generative AI Permission Level

Assistance with Attribution

This assessment requires you to write/create a first iteration of your submission yourself. You are then permitted to use generative AI tools, software or services to improve your submission in the ways set out below.

Any output of generative AI tools, software or services that is used within your assessment must be attributed with full referencing.

If outputs of generative AI tools, software or services form part of your submission and are not appropriately attributed, your Convenor will determine whether the omission is significant. If so, you may be asked to explain your submission. If you are unable to satisfactorily demonstrate your understanding of your submission you may be referred to UNSW Conduct & Integrity Office for investigation for academic misconduct and possible penalties.

For more information on Generative AI and permitted use please see [here, opens in a new window](#).

An AI fundamental training module is recommended as a course activity prior to the literature review assessment. This can be accessed via the "AI" menu at the topmost of the Moodle page, and selecting "Be Ready with Generative AI" or [this direct link](#).

Test 1

Assessment Overview

This assessment is timetabled in Week 7.

The test is 45 minutes and comprises multiple choice questions. The test covers course content delivered in the first half of the course (Weeks 1 to 5 inclusive). You will receive feedback on how you performed for each topic on Moodle.

Course Learning Outcomes

- CL02 : Identify and compare different cell types and their structures and functions.
- CL03 : Describe the structures of the macromolecules proteins, fats and carbohydrates, and their metabolism including the processes for energy generation in cells.
- CL05 : Demonstrate a knowledge of the techniques and safe work practices required in a biological laboratory, including use of the light microscope, related calculations and basic

molecular biology methods.

Detailed Assessment Description

See the Assessment Hub on Moodle for more details.

The assessment will take place during your scheduled laboratory time. The test will commence at the start of the class and last for 45 minutes. Students must be present at the start of the class to ensure they have the full 45 minutes to complete the test.

Students who apply for Special Consideration should expect to have a supplementary test scheduled during a timetabled lecture time in Week 10 or in Week 11.

Submission notes

The test will be done online and automatically submits when the test time ends if not submitted prior.

Assignment submission Turnitin type

Not Applicable

Generative AI Permission Level

No Assistance

This assessment is designed for you to complete without the use of any generative AI. You are not permitted to use any generative AI tools, software or service to search for or generate information or answers.

For more information on Generative AI and permitted use please see [here, opens in a new window](#).

Science Communication Project

Assessment Overview

In Week 9 you will deliver a short (approx. 3 min) presentation on the topic of your literature review. This will be supported by a visual abstract, that is an annotated visual summary of your literature review/presentation topic. You will be expected to answer some questions following the talk and engage in asking questions to other presenters.

The visual abstract contributes 10%, the presentation 10% and the question time 5%.

Support for this assessment is provided as a lecture on Science Communication and detailed instructions provided in the Assessment Hub on Moodle.

This assignment is due in Week 9.

Written feedback and a completed rubric is provided.

Course Learning Outcomes

- CLO1 : Communicate findings from scientific literature to different audiences using verbal and written formats.

Detailed Assessment Description

See the Assessment Hub on Moodle for more details including the Instructions and Marking Rubric.

Note that students with laboratory times that clash with a public holiday will be provided information on their alternative times in class.

Submission notes

Visual abstract submitted via Turnitin and presentations delivered during class.

Assessment information

An AI fundamental training module is recommended as a course activity prior to the literature review assessment. This can be accessed via the "AI" menu at the topmost of the Moodle page, and selecting "Be Ready with Generative AI" or [this direct link](#).

Students impacted by public holidays will be provided details of alternative arrangements.

Assignment submission Turnitin type

This assignment is submitted through Turnitin and students do not see Turnitin similarity reports.

Generative AI Permission Level

Assistance with Attribution

This assessment requires you to write/create a first iteration of your submission yourself. You are then permitted to use generative AI tools, software or services to improve your submission in the ways set out below.

Any output of generative AI tools, software or services that is used within your assessment must be attributed with full referencing.

If outputs of generative AI tools, software or services form part of your submission and are not appropriately attributed, your Convenor will determine whether the omission is significant. If so, you may be asked to explain your submission. If you are unable to satisfactorily demonstrate

your understanding of your submission you may be referred to UNSW Conduct & Integrity Office for investigation for academic misconduct and possible penalties.

For more information on Generative AI and permitted use please see [here, opens in a new window](#).

Students who apply for Special Consideration and are approved should expect to present during a timeslot usually used for lectures in Week 11.

An AI fundamental training module is recommended as a course activity prior to the project assessment. This can be accessed via the "AI" menu at the topmost of the Moodle page, and selecting "Be Ready with Generative AI" or [this direct link](#).

Test 2

Assessment Overview

This assessment is timetabled in Week 10.

The test is 45 minutes and comprises multiple-choice questions. The test covers course content delivered in the second half of the course (Weeks 7 to 9 inclusive). You will receive feedback on how you performed for each topic on Moodle.

Course Learning Outcomes

- CL04 : Describe the structures of genetic material, the processes involved in cell division and gene expression, and how this applies to genetic inheritance and evolution.
- CL05 : Demonstrate a knowledge of the techniques and safe work practices required in a biological laboratory, including use of the light microscope, related calculations and basic molecular biology methods.

Detailed Assessment Description

See the Assessment Hub on Moodle for more details.

The assessment will take place during your scheduled laboratory time. The test will commence at the start of the class and last for 45 minutes. Students must be present at the start of the class to ensure they have the full 45 minutes to complete the test.

Note that students with laboratory times that clash with a public holiday will be provided information on their alternative times in class.

Students who apply for Special Consideration should expect to have a supplementary test scheduled during a timetabled lecture time in Week 10 or in Week 11.

Submission notes

The test will be done online and automatically submits when the test time ends if not submitted prior.

Assessment information

Students impacted by public holidays will be provided details of alternative arrangements.

Assignment submission Turnitin type

Not Applicable

Generative AI Permission Level

No Assistance

This assessment is designed for you to complete without the use of any generative AI. You are not permitted to use any generative AI tools, software or service to search for or generate information or answers.

For more information on Generative AI and permitted use please see [here, opens in a new window](#).

General Assessment Information

Further information on the assessments is provided in the Assessment Hub on Moodle. If you have any questions, please attend one of our Q&A sessions, post on the Moodle Forum or email BABS1201@unsw.edu.au.

Grading Basis

Standard

Requirements to pass course

To pass this course you need a composite mark of at least 50.

Course Schedule

Teaching Week/Module	Activity Type	Content
Week 1 : 15 September - 21 September	Lecture	Welcome to BABS1201 - Hassan Choucair and Rebecca LeBard
	Lecture	Scientific literature - Rebecca LeBard (pre-recorded on Moodle)
	Lecture	Science communication - Hassan Choucair (pre-recorded on Moodle)
	Laboratory	Health and Safety Please come prepared having: (1) set up your Class Notebook on Teams (2) purchased your lab coat and safety glasses and (3) located the laboratory location found on your timetable (Teaching lab 1).
Week 2 : 22 September - 28 September	Lecture	Lecture 3: Life (pre-recorded) Rebecca LeBard
	Lecture	Lecture 4: Cells I (pre-recorded) Rebecca LeBard
	Lecture	Lecture 5: Cells II (pre-recorded) Rebecca LeBard
	Laboratory	Cell structure
Week 3 : 29 September - 5 October	Lecture	Lecture 6: Macromolecules I (pre-recorded) Rebecca LeBard
	Lecture	Lecture 7: Macromolecules II (pre-recorded) Rebecca LeBard
	Lecture	Lecture 7: Macromolecules III (pre-recorded) Rebecca LeBard
	Laboratory	Macromolecules
Week 4 : 6 October - 12 October	Lecture	Lecture 9: Cell integrity (pre-recorded) Hassan Choucair
	Lecture	Lecture 10: Nutrient and ion transport (pre-recorded) Hassan Choucair
	Lecture	Lecture 11: Metabolism I Rebecca LeBard
	Laboratory	Osmosis and diffusion
Week 5 : 13 October - 19 October	Lecture	Lecture 12: Metabolism II (pre-recorded) Rebecca LeBard
	Lecture	Lecture 13: Photosynthesis (pre-recorded) Rebecca LeBard
	Laboratory	Photosynthesis
Week 7 : 27 October - 2 November	Lecture	Lecture 14: Science communication 2 Hassan Choucair
	Lecture	Lecture 15: DNA replication (pre-recorded) Teagan Mock
	Lecture	Lecture 16: Cell division and reproduction (pre-recorded) Teagan Mock
	Lecture	Lecture 17: Gene expression I (pre-recorded) Teagan Mock
	Laboratory	Mitosis and cell division
	Assessment	Test 1
Week 8 : 3 November - 9 November	Lecture	Lecture 18: Gene expression II (pre-recorded) Teagan Mock
	Lecture	Lecture 19: Polymerase chain reaction (pre-recorded) Teagan Mock
	Lecture	Lecture 20: Mutation (pre-recorded) Teagan Mock
	Laboratory	Genetic inheritance
Week 9 : 10 November - 16 November	Lecture	Lecture 21: Mendel's laws of heredity (pre-recorded) Teagan Mock
	Lecture	Lecture 22: Mechanisms of inheritance (pre-recorded) Teagan Mock
	Lecture	Lecture 23: Population genetics (pre-recorded) Teagan Mock
	Assessment	Presentations (in Laboratory class time)

Attendance Requirements

Students are strongly encouraged to attend all classes and review lecture recordings.

General Schedule Information

Your attendance is expected for all laboratory classes. If you are unable to attend a laboratory class, please email BABS1201@unsw.edu.au providing the reasons you need a make-up activity.

For laboratory classes, students are required to:

- Arrive on time
- Be prepared for the class (correct PPE of laboratory coat, safety glasses and closed shoes, and a device to access the online Class Notebook)
- Actively participate in the activities.

If you require a device to use in class at any time, please borrow one from the UNSW Library:

<https://www.library.unsw.edu.au/using-the-library/computers#loanlaptops>

Course Resources

Prescribed Resources

There is no laboratory manual for purchase. The laboratory classes use the Class Notebook (OneNote) which is introduced in the first lecture and accessible via the course Teams site.

Recommended Resources

Lisa A. Urry, Noel Meyers, Michael L. Cain, Steven A. Wasserman, Peter V. Minorsky. 12th edition, **Campbell Biology Australian and New Zealand version**. This is available in either print <https://www.bookshop.unsw.edu.au/details.cgi?ITEMNO=9781488626241>, opens in a new window and electronic format <https://unswbookshop.vitalsource.com/products/-v9781488626272>, opens in a new window

Copies are also available for borrowing in the UNSW Library (both print and electronic).

Additional Costs

A laboratory coat and safety glasses are required for this course.

Course Evaluation and Development

Student feedback on the course is collected in both Terms 1 and 3 iterations of the course via myExperience. Changes made to the course are reported back to students via myFeedback matter in the Quick Links section of the course Moodle page.

Staff Details

Position	Name	Email	Location	Phone	Availability	Equitable Learning Services Contact	Primary Contact
Convenor	Rebecca LeBard				Available by appointment and during scheduled Q&A times as advertised in the course announcements.	No	No
	Hassan Choucair				Available by appointment and during scheduled Q&A times as advertised in the course announcements.	Yes	Yes

Other Useful Information

Academic Information

Upon your enrolment at UNSW, you share responsibility with us for maintaining a safe, harmonious and tolerant University environment.

You are required to:

- Comply with the University's conditions of enrolment.
- Act responsibly, ethically, safely and with integrity.
- Observe standards of equity and respect in dealing with every member of the UNSW community.
- Engage in lawful behaviour.
- Use and care for University resources in a responsible and appropriate manner.
- Maintain the University's reputation and good standing.

For more information, visit the [UNSW Student Code of Conduct Website](#).

Academic Honesty and Plagiarism

Referencing is a way of acknowledging the sources of information that you use to research your assignments. You need to provide a reference whenever you draw on someone else's words, ideas or research. Not referencing other people's work can constitute plagiarism.

Further information about referencing styles can be located at <https://student.unsw.edu.au/referencing>

Academic integrity is fundamental to success at university. Academic integrity can be defined as a commitment to six fundamental values in academic pursuits: honesty, trust, fairness, respect, responsibility and courage. At UNSW, this means that your work must be your own, and others' ideas should be appropriately acknowledged. If you don't follow these rules, plagiarism may be detected in your work.

Further information about academic integrity, plagiarism and the use of AI in assessments can be located at:

- The [Current Students site](#),
- The [ELISE training site](#), and
- The [Use of AI for assessments](#) site.

The Student Conduct and Integrity Unit provides further resources to assist you to understand your conduct obligations as a student: <https://student.unsw.edu.au/conduct>

Submission of Assessment Tasks

Penalty for Late Submissions

UNSW has a standard late submission penalty of:

- 5% per day,
- for all assessments where a penalty applies,
- capped at five days (120 hours) from the assessment deadline, after which a student cannot submit an assessment, and
- no permitted variation.

Any variations to the above will be explicitly stated in the Course Outline for a given course or assessment task.

Students are expected to manage their time to meet deadlines and to request extensions as early as possible before the deadline.

Special Consideration

If circumstances prevent you from attending/completing an assessment task, you must officially apply for special consideration, usually within 3 days of the sitting date/due date. You can apply by logging onto myUNSW and following the link in the My Student Profile Tab. Medical documentation or other documentation explaining your absence must be submitted with your application. Once your application has been assessed, you will be contacted via your student email address to be advised of the official outcome and any actions that need to be taken from there. For more information about special consideration, please visit: <https://>

Important note: UNSW has a “fit to sit/submit” rule, which means that if you sit an exam or submit a piece of assessment, you are declaring yourself fit to do so and cannot later apply for Special Consideration. This is to ensure that if you feel unwell or are faced with significant circumstances beyond your control that affect your ability to study, you do not sit an examination or submit an assessment that does not reflect your best performance. Instead, you should apply for Special Consideration as soon as you realise you are not well enough or are otherwise unable to sit or submit an assessment.

Faculty-specific Information

Additional support for students

- [The Current Students Gateway](#)
- [Student Support](#)
- [Academic Skills and Support](#)
- [Student Wellbeing, Health and Safety](#)
- [Equitable Learning Services](#)
- [UNSW IT Service Centre](#)
- Science EDI Student [Initiatives](#), [Offerings](#) and [Guidelines](#)